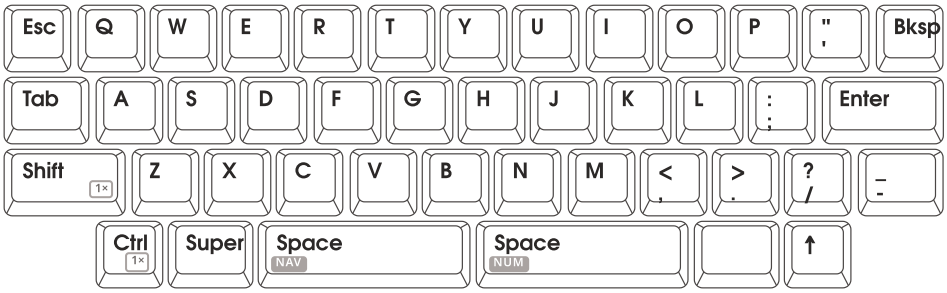
Agar mini

Custom ZMK Layout
Key Function Manual

ZMK FIRMWARE

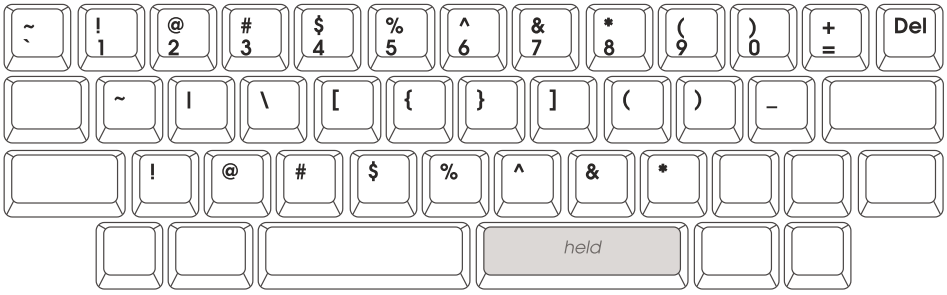
Quick tip: a **grey tag** is what the key does when held. **1×** marks a sticky key: tap it, then the next key. Blank keys pass through to Layer 0.

Layer 0 BASE



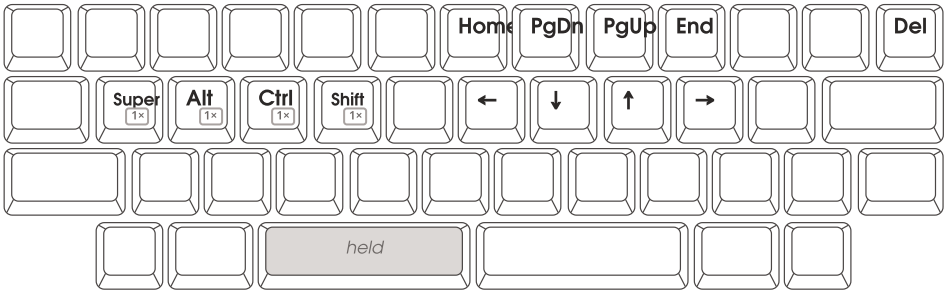
- 1× **Sticky:** tap, then the next key
- ' **Accent key:** tap it, then the vowel (Layer 4), or Space for '; Shift gives "
- NUM **Hold for Layer 1**, right Space
- NAV **Hold for Layer 2**, left Space
- **Both Spaces** for Layer 3
- Super **Plain key**, hold it
- Shift **The only Shift**, bottom-left as usual
- ↑ **Up arrow** for the last command

Layer 1 NUM



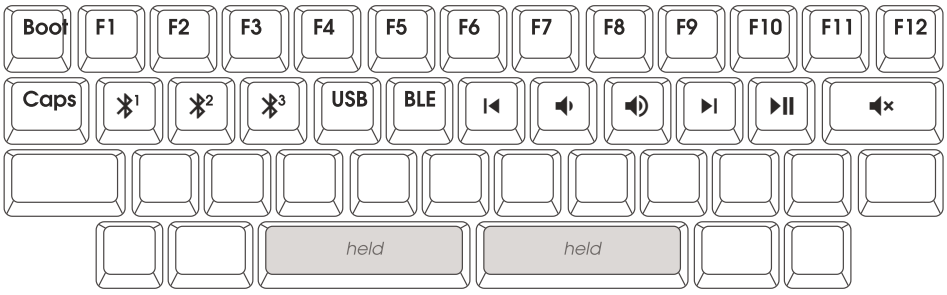
- **Hold right Space**
- 1–0 **Super + number** switches mango workspace
- . / - **Pass through** for ~/, 1.5, IPs
- ` ~ ^ **Plain characters**, not dead keys

Layer 2 NAV



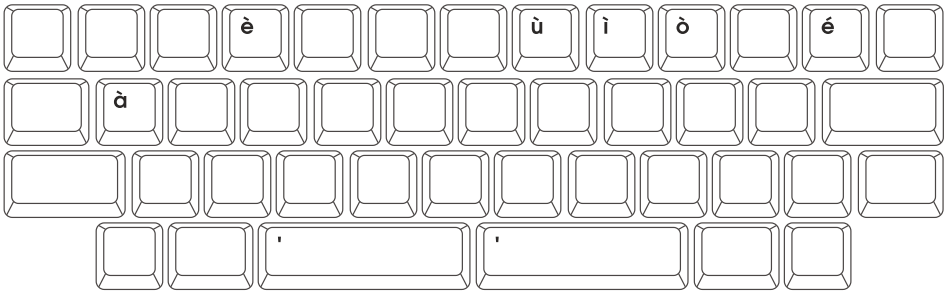
- **Hold left Space**
- ← ↓ ↑ → **Arrows on hjkl**
- 1× **Sticky modifiers:** tap D, tap L = Ctrl+→
- Home **Home, PgDn, PgUp, End** sit above the arrows

Layer 3 FN



- **Hold both Spaces**
- BT1 **Bluetooth profile 1–3**; hold 3 s to clear its pairing
- USB **Switch output** to USB or BLE
- Boot **Bootloader:** hold 3 s; a tap restarts
- ⏮ ⏯ ⏭ **Media** previous, play/pause, next
- 🔉 🔊 🔇 **Volume** down, up, mute

Layer 4 ACC



- 1× **Tap '**, then the key; expires after 1.5 s
- ' **Either Space** types the apostrophe; ' twice types é
- È **Capital:** tap Shift after '
- . **Other keys pass through**; the apostrophe is lost

Everyday actions

ACTION	SENDS	KEYS
è à ì ò ù		tap [']`, tap the vowel
é		tap [']` twice
È		tap [']`, tap [Shift], tap [E]
Backtick	`` ` ``	hold [R Space] + [Esc]
Capital letter		tap [Shift] (bottom left), tap the letter
Apostrophe / quote	`' / "`	tap [']`, tap [Space] / tap [Shift], tap [']`
Newline in Claude Code	`Shift+Enter`	tap [Shift], tap [Enter]
Question / exclamation mark	`? / !`	tap [Shift], tap [/] / hold [R Space] + [Z]
Number	`1 … 0`	hold [R Space] + [Q] ... [P]
Backspace		top-right corner, as on the Q4
Run the last command again	`Up, Enter`	tap [↑] (bottom right), tap [Enter]
Copy / paste	`Ctrl+C / Ctrl+V`	tap [Ctrl], tap [C] / [V]
Kitty copy / paste	`Ctrl+Shift+C / V`	tap [Ctrl], tap [Shift], tap [C] / [V]
Switch workspace	`Super+3`	hold [Super] + hold [R Space] + [E]
Move window to workspace	`Super+Shift+3`	tap [Shift], then hold [Super] + hold [R Space] + [E]
Other mango shortcuts	`Super+Return, Super+q`	hold [Super] + the key
Focus direction	`Super+Left`	hold [L Space], tap [A], tap [H]
Kitty tab	`Alt+3`	hold [L Space], tap [S] (Alt), then hold [R Space] + [E]
Accept fish autosuggestion	`Right`	hold [L Space] + [L]

No mango config changes are needed: every action sends exactly what a full-size keyboard would.

Timing

KEY	FLAVOR	TERM	QUICK-TAP	PRIOR IDLE
Space / NAV	balanced	200	175	150
Space / NUM	balanced	200	175	—

STICKY KEYS	RELEASE AFTER	QUICK RELEASE
Shift, Ctrl, Alt, Super	1500	yes
' (sticky layer)	1500	no

- All values in milliseconds. Letters have no hold behaviour: they type on press.
- Sticky keys apply to the next key only and expire after 1.5 s. Tapping Ctrl then Shift stacks them. They can also be held normally. `'` ignores modifiers, so Shift tapped between `'` and the vowel still gives the capital; Shift before `'` gives `"`. A key other than a vowel, `'` or Space drops the apostrophe.
- Space / NUM has no prior-idle setting; otherwise typing `port 8080` quickly would give letters instead of digits.

Design

1. **Thumbs hold, nails tap one key at a time.** No chords; ring fingers only reach Esc and Backspace, the pinky only Shift, with the back of the finger.
2. **`'` is a dead key**, as on `us intl`: `'` + vowel = è, `'` + Space = `'`, with Italian grave accents.
3. **Sticky Shift, Ctrl and Alt:** tap, then the next key.
4. **`" ` ~ ^` are plain**, despite `us intl` dead keys.
5. **Shift where it always is,** bottom-left. Super under the left thumb; the right thumb key is free. Backspace stays top-right.

Requires `us intl`

Accents and plain quotes are typed as `us intl` dead-key sequences: `` ` `` + vowel for è, `'` + `e` for é, `'` + Space for `'`. On a device with a plain US layout, accents come out as `` `e `` and quotes get an extra space.

Dropped from stock

Mouse keys (trackpad available), Win+V, Print Screen (mango uses Super+Shift+S), Insert, Pause, Scroll Lock, right Alt, arrow cluster (↑ kept bottom right, the rest on NAV).